Venkata Lakshmi Parimala Pasupuleti

Software Engineer, Agents · Agent Design · Eval Harnesses · Prompt Engineering · Context Management · Model Selection · Observability · Python · LLM APIs

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SUMMARY

Software engineer specializing in **production AI agent design, eval harnesses, prompt engineering, and observability** - building agents that actually get the job done. Currently at Intuit: **agent architectures** with tool calling, context window management, and prompt strategies for Anthropic Claude in production financial services; **eval frameworks** with LLM-as-judge scoring, golden datasets, and CI regression gating; **observability** via structured logging and production quality signals; iterative improvement cycles turning failure analysis into eval coverage. Built **Leasel** - a production domain-specific agent platform: 8-agent LangGraph orchestration with skill writing, optimal tool design, MCP server integration, context management strategies, HITL guardrails, and a CI eval harness (93.6% precision, 1,240 test cases). Built **AgentProbe** - an open-source agent eval and observability platform with **model selection** across 6 providers (cost-aware routing: Anthropic, OpenAI, Azure OpenAI, AWS Bedrock, Groq, Ollama), CI regression gating, per-run tracing, and structured instrumentation. At Entrata (7+ years, 100K+ users), owned mission-critical async distributed systems in regulated multi-tenant SaaS. Python, FastAPI, PostgreSQL, AWS, Docker. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

Domain-Specific Agent Design · Agent Environments · Actions · Skill Writing
Eval Harnesses · Iteration Loops · Task Completion Quality · LLM-as-Judge
Prompt Engineering · Context Window Management · Structured Outputs
Model Selection · Multi-Provider Routing · Cost-Aware Optimization
Tool Design · Function Calling · MCP · Parallel Tool Calls
Observability · Agent Behavior Profiling · Tracing · Instrumentation
Guardrails · HITL · Failure Handling · Production Reliability
LangGraph · LangChain · Multi-Agent Orchestration · Subagent Patterns
Golden Datasets · CI Regression · Proof-of-Concept Iteration
Python · FastAPI · TypeScript · LLM APIs · Anthropic · OpenAI
AWS · Docker · Kubernetes · PostgreSQL · CI/CD
End-to-End Ownership · Ship v0s · Cross-Functional Collaboration

EXPERIENCE

Software Engineer

Jun 2026 - Present

Intuit — Virginia, USA (Remote)

- **Agent design and prompt engineering in production:** Build LangChain-based agent architectures integrating Anthropic Claude - tool/function calling with optimal tool design, context window management strategies, structured outputs, and prompt engineering for multi-step financial workflows; iterate rapidly from proof-of-concept to v0 to production.
- **Eval harnesses and observability:** Build LLM-as-judge eval frameworks with golden datasets and CI regression gates to drive faster iteration loops; profile agent behavior via structured logging and CloudWatch instrumentation; identify bottlenecks through qualitative failure analysis and convert to eval coverage. Maintain guardrails and HITL for safety-critical decisions.
- Build ML data pipelines (scikit-learn, XGBoost); React/TypeScript front ends; deploy via Docker + GitHub Actions CI/CD to AWS.

Senior Software Engineer

Oct 2020 - Jul 2024

Entrata — Affordable Housing SaaS (regulated multi-tenant, 100K+ monthly active users)

- Owned mission-critical data pipelines and async distributed systems end-to-end - HUD compliance ETL (180K records, zero violations across 6 years), RabbitMQ event-driven processing with idempotency guarantees, PHI-compliant AWS S3 migration (100% coverage, zero incidents).
- Led Docker microservices redesign (50% scalability improvement); optimized 200+ SQL queries (30% performance gain); Python/Bash automation eliminating 80% of manual workload; mentored 5+ engineers; 300+ code reviews across 120+ Agile sprints.

Junior Software Engineer

Sep 2018 - Oct 2020

Entrata India — Pune

- Built backend modules (Python, PHP, Java, PostgreSQL); delivered large-scale data integration workflows (1.6M records); led AWS S3 migration from legacy storage.

PROJECTS

Leasel - Domain-Specific Agent Platform: Skill Writing · Tool Design · Context Mgmt · Evals

2026 | Python, LangGraph (LangChain), FastAPI, ChromaDB, MCP, CI/CD | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Production domain-specific agent platform: **8-agent LangGraph orchestration** with skill writing, optimal tool design, function calling, and conditional routing; **MCP server integration** for shared tool access; **context window management** with durable state and retrieval-grounded context; **HITL guardrails** for regulated decisions; **CI eval harness** with LLM-as-judge, golden datasets, and iteration-loop optimization: 93.6% precision, 87.9% F1 across 1,240 test cases. Iterated from proof-of-concept through production across three product surfaces.

AgentProbe - Agent Eval, Model Selection & Observability Platform

2025 | Python, FastAPI, PostgreSQL, CI/CD | github.com/venkatalakshmiparimala/AgentProbe

- Open-source agent eval and observability platform: offline golden-dataset evals, LLM-as-judge scoring, CI regression harness for faster iteration loops, failure handling with structured error classification; **cost-aware model selection** across Anthropic, OpenAI, Azure OpenAI, AWS Bedrock, Groq, and Ollama; full per-run tracing and structured instrumentation for agent behavior profiling.

AI on FHIR — Domain-Specific Clinical AI & Tool Integration

2024 | Python, Flask, React, BERT, LLMs, FHIR REST APIs, Docker | github.com/venkatalakshmiparimala/AI_on_FHIR

- Translates natural-language clinical questions into FHIR REST API calls using LLMs, BERT, and embeddings - cut clinical lookup time ~60%. Domain-specific tool design, OAuth-secured enterprise API integration, HIPAA-style RBAC, Docker-containerized.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia I GPA 3.87 May 2026

B.Tech Computer Science & Engineering — JNTU Anantapur 2017